Solutions

1).

(a) *P*(indicates analyzing data as critical to his or her job) = 579/746 = 0.78.
(b) *P*(is a manager) = 336/746 = 0.45
(c) *P*(indicates analyzing data as critical to his or her job or is a manager)
= 579/746 + 336/746 - 255/746 = 0.88
(d) The probability of saying that analyzing data is critical or is a manager includes the probability of saying that analyzing data is critical plus the probability of being a manager minus the joint probability of saying that analyzing data is critical and is a manager.

2)

(a) *P*(prefers name-brand) = 660/2276 = 0.29
(b) *P*(is male and prefers name-brand) = 319/2276 = 0.14
(c) *P*(is male or prefers name-brand) = (1138+660 –319)/2276 = 0.65
(d) The probability of "is male or prefers name-brand" includes the probability of "is a male", the probability of "prefers name-brand", minus the joint probability of "is male and prefers name-brand".

3)

Sample Space		Gender		
		Male	Female	Totals
Influenced	Yes	89	173	262
	No	761	677	1438
	Totals	850	850	1700

(a) *P*(social media does influence his/her purchase decisions) = 262/1700 = 0.15
(b) *P*(female and is a person whose purchase decisions are influenced by social media) = 173/1700 = 0.10
(c) *P*(female or is a person whose purchase decisions are influenced by social media) = (850+262-173/1700 = 0.55
(d) *P*(male *or* female) = 1700/1700 = 1.00

4).

(a) *P*(Increased use of LinkedIn | B2B marketer) = 912 /1,184 = 0.77
(b) *P*(Increased use of LinkedIn | B2C marketer) = 920/1,703 = 0.54
(c) P(Increased use of LinkedIn) = 0.63, which is not equal to P(Increased use of Linked | - B2B marketer) = 0.77. Therefore, increased use of LinkedIn and business focus are not independent.

5).

(a) P(he prefers name-brand | male) = 319/1,138 = 0.28
(b) P(she prefers name-brand | female) = 341/1,138 = 0.30
(c) Since P(prefers name-brand) = 0.29 is not equal to P(prefers name-brand | male) = 0.28, preference and gender are not statistically independent.

6)

(a) P(analyzing data is critical | is staff) = 324/410 = 0.79
(b) P(analyzing data is not critical | is staff) = 86/410 = 0.21
(c) P(analyzing data is critical | is a manager) = 255/336 = 0.76
(d) P(analyzing data is not critical | is a manager) = 81/336 = 0.24

7).

S_1 = successful, S_2 = not successful and B = request received for additional information.

a. $P(S_1) = .50$

b. $P(B \mid S_1) = .75$

c. $$P(S_1 \mid B) = \frac{(.50)(.75)}{(.50)(.75)+(.50)(.40)} = \frac{.375}{.575} = .65$$

8)

M = missed payment
D_1 = customer defaults
D_2 = customer does not default

$P(D_1) = .05 \quad P(D_2) = .95 \quad P(M \mid D_2) = .2 \quad P(M \mid D_1) = 1$

a. $$P(D_1 \mid M) = \frac{P(D_1)P(M \mid D_1)}{P(D_1)P(M \mid D_1)+P(D_2)P(M \mid D_2)} = \frac{(.05)(1)}{(.05)(1)+(.95)(.2)} = \frac{.05}{.24} = .21$$

b. Yes, the probability of default is greater than .20.

9)

a. Let M = event that a putt is made

$P(M)$ = 983,764/1,613,234 = .610

Note: The probability that a putt is missed is $P(M^C) = 1 - P(M) = 1 - .610 = .390$

b. Let *A*= event that a PGA Tour player has a par putt

$$P(A|M) = .640$$

$$P(A|M^C) = .203$$

$$P(M|A) = \frac{P(M)P(A|M)}{P(M)P(A|M) + P(M^C)P(A|M^C)} = \frac{(.610)(.640)}{(.610)(.640) + (.390)(.203)} = .831$$

c. Let *B* = event that a PGA Tour player has a birdie putt

$$P(B|M) = .188$$

$$P(B|M^C) = .734$$

$$P(M|B) = \frac{P(M)P(B|M)}{P(M)P(B|M) + P(M^C)P(B|M^C)} = \frac{(.610)(.188)}{(.610)(.188) + (.390)(.734)} = .286$$

d. These probabilities indicate that there is a much higher probability of making a par put than a birdie putt. The authors of the article referenced in this exercise state that "even the best players ... show evidence of loss aversion." In other words, PGA Tour players tend to view missing a par putt as a loss and making a birdie putt as a gain. With regard to putting, their behavior indicates that they prefer avoiding losses to making gains. "On average this bias costs the best golfers over $1.2 million in tournament winnings per year."